Lab 3: To implement Logistic Regression for binary or multiclass classification problems and analyse the classification performance.

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```
import seaborn as sns
import pandas as pd
import numpy as np
```

```
df = sns.load_dataset('iris')
df.head()
```

```
df.describe()
```

```
df.info()
```

```
df.shape
```

```
df['species'].unique()
```

```
df.isnull().sum()
```

```
df=df[df['species']!='setosa']
```

```
df.shape
```

```
df['species'].unique()
```

```
df['species'] = df['species'].map({'versicolor': 0, 'virginica': 1})
```

```
X = df.iloc[:,:-1]
print(X)
```

```
y = df.iloc[:,-1]
print(y)
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=0.7,test_size=0.3,random_state=42)
```

```
print(X_train.shape)
```

```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression()
```

```
# Hyper parameter tunning
from inspect import Parameter
from sklearn.model_selection import GridSearchCV
parameter={'penalty':['l1','l2','elasticnet'],'C':[1,2,3,4,5,6,10,20,30,40,50],'max_iter':[100,200,300,400]}
```

```
classifier_regressor = GridSearchCV(classifier, param_grid= parameter, scoring='accuracy',cv=5)
```

```
classifier_regressor.fit(X_train,y_train)
```

```
print(classifier_regressor.best_params_)
```

```
print(classifier_regressor.best_score_)
```

```
y_pred = classifier_regressor.predict(X_test)
```

```
from sklearn.metrics import accuracy_score, classification_report
```

```
score = accuracy_score(y_pred, y_test)
print(score)
```

```
# For Precision,  Recall, F1 Score & Accuracy
print(classification_report(y_pred, y_test))
```

```
sns.pairplot(df, hue='species')
```

```
df.corr()
```